Exact Pareto Corners and Uniform State–Stack Tradeoffs in a Binary Deterministic Pushdown Hierarchy

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Abstract

We examine the joint state and stack-alphabet resources necessary to recognize a family $B_{m,k}$ of binary deterministic context-free languages. The model permits arbitrary finite stack replacements and deterministic ε-moves but starts with one physical stack letter and accepts after complete input by final control and a physically empty stack. For every $m \geq 2$, $r \geq 1$, and $g \geq mr$, we identify a genuine Pareto-minimal resource point (mr, g) for $B_{m,r(g-1)}$. More generally, we determine exact stack-budget-constrained state thresholds in an explicit interval of parameters. We also establish a two-sided uniform estimate for the cardinality of the entire Pareto frontier, of order $\Theta(\sqrt{mk})$ with absolute constants independent of m, k. On a square-root-shifted two-parameter region, every state budget $p = 1, \ldots, m$ is represented by a distinct Pareto point, and the parameter threshold is exact for the associated product-bound/compiler-bound certificate. The exact full frontier remains open. General state–stack alphabet recodings are known; our focus is on exact resource coordinates and frontier width for the particular binary witness family. The main results concern the unrestricted deterministic ε-move model; they do not provide an exact formula for every Pareto coordinate.

Keywords: deterministic pushdown automata; Pareto frontier; stack alphabet; state complexity; descriptional complexity; epsilon transitions.
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1 Introduction and relation to prior work

State complexity and the stack alphabet are independent descriptional resources in a pushdown automaton. Their tradeoff has a long history. General state reduction can entail the loss of deterministic behavior [2]; reducing stack alphabets by increasing state sets has also been studied [4, 5]. The fixed-state hierarchy of Masopust [6] addresses a distinct initial-store setting. Kiss and Rovan [3] consider descriptional bounds, particularly the extreme subfamilies of one-state or two-stack-symbol PDAs, and point out difficulties with a scalar complexity measure combining the two resources. We instead retain the coordinatewise partial order and ask for exact minimal pairs and the cardinality of the resulting frontier for one explicit *binary deterministic* family. The older works use varying acceptance modes and degrees of determinism, so numerical results cannot simply be transferred between their conventions and ours. A general recoding is not presented as a new discovery.

The witness family and representation-independent product bound originate in an earlier work by the author, deposited on Zenodo [1]. That work addresses the binary fixed-state stack-alphabet hierarchy; the present paper examines simultaneous resource minima, exact corner pairs and uniform Pareto width. The definition of the witness and the main arguments needed here are included, so no unavailable result is invoked merely by citation.

There are three complementary results. First, explicit machines attain infinitely many resource pairs that cannot be coordinatewise improved by *any* equivalent deterministic ε-PDA. Second, the number of minimal pairs in the entire frontier is uniformly of square-root order when m and

k vary freely. Third, we certify that every state coordinate up to m is Pareto-supported in an explicit square-root-shifted parameter region. Exact coordinate values, state-coordinate support, and cardinality are different kinds of information.

Main contributions. For the specified physical acceptance model, we (i) derive a representation-independent resource-product obstruction, the unary-stack impossibility result, and the exact one-control stack-alphabet threshold $mk + \min(m, 3)$; (ii) combine explicit all-input recodings with the lower bound to obtain exact state thresholds on parameter strips and genuine divisor-corner Pareto points; (iii) establish uniform upper and lower bounds showing that the number of Pareto points is $\Theta(\sqrt{mk})$; and (iv) prove full support of the state coordinates $1, \ldots, m$ on an explicit parameter region. The stated boundary of that region is sharp for the particular integer-inequality certificate used here, not asserted to be the earliest possible boundary for actual Pareto support. These results do not determine all resource-coordinate pairs.

Scope of the results. The binary witness and its product lower bound are stated and proved below, as are the resource recoding and the quantitative consequences. The stronger product bound $qg \geq mk + 2$, which is not required for our theorems, is not assumed. Exact state-threshold equalities, individual Pareto-minimal points and complete Pareto-frontier classifications are kept distinct throughout.

2 Model and the binary witness

A deterministic pushdown automaton is $(Q, \{0, 1\}, \Gamma, \delta, q_0, Z_0, F)$ with one initial stack symbol $Z_0 \in \Gamma$. Its transition function is a partial map

$$\delta : Q \times \Gamma \times (\{0, 1\} \cup \{\varepsilon\}) \rightharpoonup Q \times \Gamma^*.$$

For each state and top symbol, an enabled ε-move excludes both real-input moves. No transition is defined from an empty physical stack. A word is accepted if and only if the entire word has been read and the machine reaches a final control state with an *empty physical stack*. There is no implicit endmarker, immortal bottom symbol, or uniform stack-replacement-length restriction.

A resource pair (q, g) is *attainable* for L if such a machine recognizes L with at most q states and at most g stack letters. Write Pareto(L) for the coordinatewise minimal attainable pairs. Set

$$b_q(L) = \min\{g : (q, g) \text{ is attainable}\}, \qquad a_g(L) = \min\{q : (q, g) \text{ is attainable}\}.$$

Only values for which the corresponding set is nonempty are used below.

Fix $m \geq 2$, $k \geq 1$ and put $\ell = mk + 1$. The canonical realtime machine $U_{m,k}$ has states $q_1, \ldots, q_m$, stack letters $A, M_1, \ldots, M_k$, initial configuration (q_1, A), and every state final. Enumerate ℓ distinct visible modes (r_j, X_j) such that $(r_1, X_1) = (q_2, A)$ and the other modes enumerate every element of $Q \times \{M_1, \ldots, M_k\}$, in the following explicit order. Let π exchange 1 and 2 in $\{1, \ldots, m\}$ and fix every other index, and put $x_{1+(a-1)k+t} = (q_{\pi(a)}, M_t)$ for $1 \leq a \leq m$ and $1 \leq t \leq k$. The enumeration ensures $r_2 = q_2$. Its essential transitions (stack top is written on the right) are

$$\begin{aligned}
\delta(q_1, A, 1) &= (q_2, X_\ell X_{\ell-1} \cdots X_1), \\
\delta(q_i, A, 0) &= (q_i, \varepsilon), \\
\delta(r_j, X_j, 0) &= (r_{j+1}, \varepsilon), \quad (1 \leq j < \ell), \\
\delta(r_\ell, X_\ell, 0) &= (q_1, \varepsilon), \\
\delta(r_j, X_j, 1) &= (r_j, A^j X_j), \quad (1 \leq j \leq \ell).
\end{aligned}$$

The only apparent overlap of zero rules is consistent. Define $B_{m,k} = L(U_{m,k})$.

Lemma 2.1 (Exact completion profiles)**.** *For $1 \leq j \leq \ell$ and $n, t \geq 0$, letting $p_j = 10^{j-1}$, we have*

$$p_j 1^n 0^t \in B_{m,k} \quad \Longleftrightarrow \quad t = N_j(n) := \ell - j + 1 + jn.$$

Proof. After p_j, the current visible symbol is X_j and $\ell - j + 1$ source-table letters remain. Each additional input 1 deposits j ordinary counter letters below the same visible mode. Subsequent zeros each remove precisely one physical symbol until the stack becomes empty. Every control state is final, so the exact completion length is the resulting height. A shorter suffix leaves a nonempty stack; a longer suffix tries to read input from the empty stack. □

3 Universal product lower bound

Theorem 3.1 (Distinct completion slopes force physical modes)**.** *Every deterministic ε-PDA recognizing $B_{m,k}$ obeys*

$$|Q|\,|\Gamma| \geq \ell = mk + 1.$$

Proof. For each j, run the unique deterministic computation along the infinite stream of ones following p_j, including every forced ε-move. This is a well-defined infinite sequence of raw control-and-entire-stack configurations: every finite prefix $p_j 1^n$ has an accepted zero completion. A raw configuration cannot recur at different consumed-one counts, for it would then accept two different pure-zero completion lengths, contradicting Lemma 2.1. Repetition at the same consumed-one count would be a forced deterministic ε-cycle: between those raw positions no real input is read, so determinism repeats the same enabled ε steps indefinitely. This is impossible along a live analysis path. Since there are only finitely many configurations at bounded physical height, the stack height tends to infinity along this path.

Consequently the path contains infinitely many *suffix-minimum-height* positions, each of which never drops to a smaller stack height at any subsequent position. Two such positions have the same physical mode (q, X); their stacks have the form ηX and $\eta\rho X$ with $\rho \neq \varepsilon$, for otherwise a full configuration would repeat. The intervening transition segment consumes 1^c with $c \geq 1$ (a purely ε segment would iterate into a forced infinite ε-computation). At no point is η inspected: every intermediate raw stack, including those reached after forced ε moves, has height at least that of the initial cut, and reaching a letter of η would require a smaller height. This remains true even when a replacement grows the stack above that cut. Therefore the identical raw transition segment can be run above *any* lower stack context, with the same consumed input symbols and the same forced ε-moves. Iteration yields a context-independent pump

$$(q, \xi X) \xRightarrow{1^{sc}} (q, \xi\rho^s X) \qquad (s \geq 0),$$

for any lower ξ.

Suppose one mode were available as such a protected pump on two different rays $i \neq j$. Choose the segment 1^c and its inserted block ρ from the first ray. Its proven independence of the buried tail replays *that same segment* from the mode occurrence on the second ray, regardless of the latter ray's own recurrence segment. Apply this pump s times at fixed occurrences on both rays. In the subsequent accepting zero-only drains, the common top block $\rho^s X$ must be erased before either fixed lower context becomes accessible. Determinism makes this erasure consume the same number of zeros and reach the same one of finitely many exit control states in both runs. At a fixed control-and-lower-stack configuration there is at most one accepted zero-only length: accepting a shorter zero word empties the stack, after which no unread zeros can be processed. Writing d_s for the common upper-block zero cost, $e_s \in Q$ for its exit control, and $\tau_e(\eta)$ for the unique accepted zero length from (e, η), the two required lengths are $d_s + \tau_{e_s}(\eta_i)$ and $d_s + \tau_{e_s}(\eta_j)$. Their difference lies in the finite set $\{\tau_e(\eta_i) - \tau_e(\eta_j) : e \in Q \text{ and both terms are defined}\}$, and is therefore bounded. Lemma 2.1 instead makes this difference $(i - j)cs + O(1)$, which is unbounded. Thus protected modes of distinct rays differ. There are ℓ rays and only $|Q||\Gamma|$ physical modes. □

Lemma 3.2 (A convenient arithmetic minimality test)**.** *Suppose every recognizer of L satisfies $qg \geq C$. If (q, g) is attainable and*

$$q(g-1) < C, \qquad (q-1)g < C,$$

then $(q, g) \in \mathrm{Pareto}(L)$.

Proof. Each strict coordinatewise improvement is dominated by one of the immediate neighbors $(q, g-1)$ or $(q-1, g)$, both forbidden by the product bound. For $q = 1$ or $g = 1$, omit the nonexistent neighbor. □

Lemma 3.3 (Unary physical stack alphabets are insufficient)**.** *For every $m \geq 2$, $k \geq 1$, no deterministic ε-PDA with one physical stack letter and finitely many states recognizes $B_{m,k}$.*

Proof. We show that $B_{m,k}$ has exponentially many distinguishable right quotients represented by words of linear length. Here use *top-first* notation just within this proof. Let $x_s = (r_s, X_s)$, $0 \leq s < \ell$, denote the zero-based canonical slope-mode list; after reading the header 1 the stack is $T = X_0 X_1 \cdots X_{\ell-1}$ in the source state r_0, and $X_0 = A$. Write

$$v_A = 0^k 101, \qquad v_B = 0^{\ell-1} 101.$$

The zero-pop, slope-one, and header rules show that either word, starting with a header T on top of an arbitrary lower stack R, re-creates that header and inserts one of the two blocks

$$W_A = A^k X_{k+1} \cdots X_{\ell-1}, \qquad W_B = A^{\ell-1}$$

beneath it, returning to the source header's post-1 state. These blocks both have length $\ell - 1$ but differ: following their first k occurrences of A, W_A has the non-A symbol $X_{k+1} = M_1$, while W_B still has A. (Since $m \geq 2$, $\ell - 1 = mk > k$.) Iterating gives, for every $b \in \{A, B\}^n$, a live prefix

$$P_b = 1 v_{b_1} \cdots v_{b_n}$$

whose source stack is $T W_{b_n} \cdots W_{b_1}$ in the same source state. All such prefixes have length at most $1 + n(\ell + 2)$ and their stacks have equal height $\ell + n(\ell - 1)$.

These stacks also have *different continuation languages.* For two different b, b', erase the common full top blocks by zeros and then erase the next k occurrences of A in their first differing block. Each complete block has the same length, and after popping it the source control has returned to the header state, so the two erased prefixes stay synchronized. The same number of zeros is read in both runs, and both source controls are in the header state. The tops are respectively A and $M_1 = X_{k+1}$; all zero erasures of complete blocks reset the header state. One following 1 increases stack height by $\ell - 1$ at the header A, whereas at M_1 it increases height by $k + 2$. If these increments coincide, supply *two* following ones instead: their total increments become ℓ and $2(k + 2)$, which cannot coincide simultaneously with $\ell - 1 = k + 2$. Therefore some common continuation of the form 1 or 11 produces different source stack heights. Supplying exactly enough zeros to empty the first of these stacks accepts on that side (every source zero rule pops one letter and all source controls are final), but not on the other. Hence the 2^n prefixes P_b have pairwise distinct right quotients.

Now suppose an equivalent deterministic ε-PDA uses only the stack letter X, with q control states and maximal replacement length $R < \infty$. Along every *live* prefix and its continuation, the forced ε-closure at a completed input boundary cannot diverge or empty the store while further input is needed. A finite forced ε-segment from positive stack height to a positive-height settled configuration has at most q steps: on a unary stack each ε-enabled control prescribes the same height increment whenever it recurs at positive height. Repeating that control would either continue the same forced cycle without a settled positive-height exit, or eventually empty the stack; neither is possible at a live boundary needing a further real input. Thus the positive-height enabled controls cannot recur before settlement. Thus after at most h consumed input symbols, the normalized unary height is bounded by

$$1 + [h + q(h + 1)] \max\{R - 1, 0\} = O(h).$$

There are at most q control states at each such height and hence only $O(h)$ possible normalized configurations. We need this estimate only for live prefixes admitting nonempty extensions: on these runs an empty stack or an infinite forced-ε suffix before the next genuine input is impossible. In a deterministic machine each normalized configuration determines one right quotient. Taking $h \leq 1 + n(\ell + 2)$ contradicts the 2^n distinct right quotients constructed above as $n \to \infty$. □

4 Constructive resource conversions

Proposition 4.1 (Typed state erasure for the witness)**.** *For $\chi = \min(m, 3)$, $B_{m,k}$ admits a one-state realtime recognizer with $d = mk + \chi$ physical stack symbols and the same singleton initial stack convention.*

Proof. Replace each source stack mode by a typed letter (q, X) recording the source state in which the letter is to be exposed. Each typed letter also has a uniquely prescribed state after its next zero-pop. Maintain the invariant that the state attached to each lower typed symbol equals the prescribed return state of the symbol directly above it. The initial letter is typed, and the long header rewrite and all slope self-rewrites preserve this return invariant. Each zero removes the top typed symbol and exposes the next one in exactly the state required by the source computation. The unique control state of the typed recognizer may therefore be fixed, with state changes encoded in these return types. This is a step-by-step simulation on all words, not a finite-word test.

There are mk marker-mode types. Among the letters of the form (q, A), only three transition roles occur: header, first-slope, and ordinary counter; the last role is absent for $m = 2$. Types sharing the ordinary-counter role have identical transitions and can be identified. The number of remaining physical letters is $mk + \min(m, 3)$, and the initial physical stack is still a single header letter. □

Theorem 4.2 (g-ary stack alphabet recoding)**.** *If a deterministic ε-PDA has n control states and $d \geq 2$ stack symbols, then for any $g \geq 2$ an equivalent machine exists with at most*

$$n \left\lceil \frac{d-1}{g-1} \right\rceil \quad \text{states and} \quad g \text{ stack symbols.}$$

The recoding preserves final-and-empty acceptance and the single physical initial stack letter, but not necessarily realtime behavior or bounded rewrite lengths.

Proof. Put $t = \lceil (d-1)/(g-1) \rceil$. Construct a prefix-free code of d source letters on g target letters using a caterpillar prefix tree with exactly t internal nodes: from each of its first $t-1$ internal nodes, expose $g-1$ leaf children and one continuing child; the last internal node has the remaining leaf children. One source letter can be assigned a one-letter root leaf labelled 0; assign it to the *actual* singleton initial source letter. Each source letter has a unique leaf code, and exactly t code prefixes need to be remembered by the finite control.

The target states are pairs (q, u), with q a source control state and u an internal tree node. At a non-leaf code continuation, a forced ε-move pops the physical code digit and changes u. At the final digit, the simulator executes the corresponding source move and replaces the decoded source symbol with the concatenated codes of the source replacement word, resetting u to the root. For right-top stack notation, stored codewords are reversed so digits are inspected in prefix order. Every configuration at a fully decoded boundary represents precisely the corresponding source configuration. At non-boundary decoder sites only an ε-move exists, so the target remains deterministic. The exact simulation preserves physical emptiness and final state at the end of input. The initial source letter's one-digit code ensures the initial physical stack has just one symbol. □

Corollary 4.3 (Universal resource bounds)**.** *Let $L = mk + 1$, $\chi = \min(m, 3)$, and $D = mk + \chi - 1$. For $p \geq 1$ and $g \geq 2$, respectively,*

$$\left\lceil \frac{L}{p} \right\rceil \leq b_p(B_{m,k}) \leq 1 + \left\lceil \frac{D}{p} \right\rceil, \tag{1}$$

$$\left\lceil \frac{L}{g} \right\rceil \leq a_g(B_{m,k}) \leq \left\lceil \frac{D}{g-1} \right\rceil. \tag{2}$$

A second bound from the canonical machine is

$$a_g(B_{m,k}) \leq m \left\lceil \frac{k}{g-1} \right\rceil.$$

Proof. The lower bounds follow from Theorem 3.1. Apply Theorem 4.2 to the one-state source from Proposition 4.1 and to the canonical $(m, k+1)$ source for the upper bounds. To obtain (1), choose the target alphabet size $g = 1 + \lceil D/p \rceil$; then $\lceil D/(g-1) \rceil \leq p$. □

5 The exact one-state endpoint

For a word u with a unique accepting zero-only continuation, let $z(u^{-1}B_{m,k})$ denote the length of that continuation; when no such continuation exists we leave z undefined. We use the same notation $z(K)$ for a language K with a unique pure-zero word.

Lemma 5.1 (Direct canonical quotient profiles)**.** *Put $B = B_{m,k}$, $\ell = mk + 1$ and $P_h = (1^{h+1})^{-1}B$ for $h \geq 0$. Then*

$$B \cap 0\{0,1\}^* = \{0\}, \qquad \varepsilon, 1, 10 \notin B,$$

and

$$\begin{aligned} z(0^{-1}P_h) &= \ell + h - 1, \\ z((10)^{-1}P_h) &= \ell + h, \\ z((10)^{-1}B) &= \ell - 1. \end{aligned}$$

For every $n \geq 1$, the unique pure-zero completion of 1^n in B has length $\ell + n - 1$.

Proof. The source initial letter A has its sole enabled zero move to the empty stack, so $0 \in B$ and no longer word beginning with 0 belongs to B. Its header move on 1 installs a stack of exactly $\ell \geq 3$ letters. Hence $\varepsilon, 1, 10 \notin B$. After 1^{h+1}, the source has taken the header move and applied the first-mode slope-one self-rewrite h times; it still exposes $X_1 = A$ in state q_2, with stack height $\ell + h$. Its zero move removes X_1 and leaves a height $\ell + h - 1$ stack that is completely drained by exactly that many zeros. If instead another 1 is read before that zero, the slope-one move inserts one additional A, yielding after the zero a height $\ell + h$ stack; it is drained by exactly $\ell + h$ zeros. These prove the first two displayed z identities. Setting $h = 0$ in the first gives the $(10)^{-1}B$ identity. The all-ones completion is the first-mode affine profile in Lemma 2.1. □

Lemma 5.2 (One-state stack concatenation)**.** *Let M be any one-state deterministic ε-PDA, with one final control state and physical-empty acceptance. If stacks are written* top first*, then for all stack words U, V,*

$$L_M(UV) = L_M(U)\, L_M(V).$$

Each nonempty $L_M(W)$ is prefix-free; in particular it contains at most one pure-zero word. If two such languages have pure-zero words, their concatenation has exactly the sum of their pure-zero lengths.

Proof. In any accepting computation from UV, the first time the symbols descending from the upper block U have all been erased separates the input into a word emptying U and a word emptying V. Until that first erasure the lower block is neither inspected nor modified, even in the presence of forced ε moves or arbitrary finite replacements. The split may occur between real inputs or during a forced ε closure; it still leaves the same sole control state and the unaltered physical lower stack. Because that state is final, both subcomputations independently accept from U and V under the same final-plus-empty convention. Conversely, they concatenate, since the lower block remains invisible until the upper block is gone. This gives equality. For prefix-freeness, an accepting computation on a shorter word empties the stack, leaving no operation available to consume additional input; determinism precludes another accepting computation from the same stack on its proper extension. □

Theorem 5.3 (Exact one-state stack threshold)**.** *For all $m \geq 2, k \geq 1$,*

$$b_1(B_{m,k}) = mk + \min(m, 3).$$

Consequently $(1, mk + \min(m, 3))$ is Pareto-minimal.

Proof. The upper bound is Proposition 4.1. For the converse let M be any one-state recognizer, with initial physical letter H. By Lemma 5.2, $L_M(H) = B := B_{m,k}$. Theorem 3.1 produces ℓ distinct protected up-mode *letters* $X_1, \ldots, X_\ell$ on the ℓ affine-slope rays; one state makes the mode-letter correspondence injective.

Every protected X_j has an accepting continuation beginning in 1. Indeed, its context-independent protected pump consumes a positive number of ones before its upper layer can be erased. Thus $\varepsilon \notin L_M(X_j)$; otherwise determinism would force the isolated X_j-layer to be erased on ε before the pump consumed input. At the live protected occurrence, an accepted continuation begins with 1. Factorization in Lemma 5.2 then forces an accepted word of $L_M(X_j)$ beginning with 1. The header H is likewise 1-live because B contains words beginning with 1.

We first show $H \neq X_j$ for all j. The required distinction concerns the physical *initial* letter, not merely a source-level header label; equality would make its full future language B available whenever that letter became exposed. On a higher-slope ray ($j \geq 2$), an occurrence of $X_j = H$ above a fixed lower stack W would give the continuation language $BL_M(W)$. The accepted zero continuations along a one-pumping run would have slope 1, since the initial 1^n cannot finish the leading B factor, and the only pure-zero completion after 1^n inside B has length $\ell + n - 1$ by Lemma 5.1; the lower contribution is constant. This contradicts the ray's slope $j \geq 2$.

For the slope-one ray, suppose its protected letter also equals H. At an occurrence after 1^{h+1} real input ones (possibly inside the forced ε-segment at that input boundary), with lower stack W, the residual language is

$$P_h = (1^{h+1})^{-1}B = BL_M(W).$$

Since B has only the zero-starting word 0, taking a left quotient by 0 yields

$$L_M(W) = 0^{-1}P_h, \qquad z(L_M(W)) = \ell + h - 1.$$

Neither 1 nor 10 is a complete word of B, so taking the left quotient by 10 in $BL_M(W)$ does not split that prefix across the two factors:

$$(10)^{-1}P_h = ((10)^{-1}B)L_M(W).$$

The right-hand side has a pure-zero word of length

$$(\ell - 1) + (\ell + h - 1) = 2\ell + h - 2.$$

Lemma 5.1, however, gives $z((10)^{-1}P_h) = \ell + h$. The difference is $\ell - 2 > 0$, so $H \neq X_1$. Altogether M requires $\ell + 1 = mk + 2$ physical letters.

For $m \geq 3$ there is one further unavoidable letter. Put $u = 10^{2k}10$. In the source machine, the first $2k$ zeros after the header select the final marker of the second group of k markers. Its input-1 rule inserts $2k + 1$ counter letters. The next zero pops the marker and enters source state q_3, exposing a counter A for which input 1 is undefined. The exposed A is a genuine zero-live but input-1-dead physical obligation. By Lemma 5.2, a top letter with an accepted continuation beginning with 1 could not represent such a stack: an accepted lower-stack suffix could be appended. The resulting stack height is exactly ℓ, so

$$u0^\ell \in B, \qquad u1v \notin B \quad (v \in \{0,1\}^*).$$

The continuation language after u is live, contains a pure-zero word, and contains no word starting in 1. The latter is a property of the entire right quotient, not merely of a convenient canonical path: every rival machine recognizing B has the same residual language after the input prefix u. At the corresponding configuration of M, let Y be the upper physical letter. By Lemma 5.2, $L_M(Y)$ is nonempty and has no word starting in 1; otherwise one could append an accepted lower-stack continuation to obtain an accepted word starting in 1. Thus Y cannot be the 1-live letter H or any X_j. It supplies an additional distinct physical letter, making at least $\ell + 2 = mk + 3$ in all $m \geq 3$ cases. Since $\min(m, 3) = 2$ for $m = 2$ and 3 otherwise, the result follows. □

6 Exact resource thresholds and Pareto corners

Theorem 6.1 (Exact state threshold strips)**.** *Fix $m \geq 2, r \geq 1, g \geq 2$. If*

$$rg - \left\lfloor \frac{g}{m} \right\rfloor \leq k \leq r(g-1),$$

then $g \geq mr$ and

$$a_g(B_{m,k}) = mr.$$

Proof. The canonical compiler from Corollary 4.3 yields $a_g \leq m\lceil k/(g-1) \rceil \leq mr$. The lower end of the interval is precisely $mk \geq (mr-1)g$; hence $(mr-1)g < mk+1$, so Theorem 3.1 gives $a_g \geq mr$. The interval is nonempty only if $\lfloor g/m \rfloor \geq r$, equivalently $g \geq mr$. □

Theorem 6.2 (Exact divisor-corner Pareto points)**.** *For any integers $m \geq 2$, $r \geq 1$ and $g \geq mr$, set $k = r(g-1)$. Then*

$$\boxed{(mr, g) \in \mathrm{Pareto}(B_{m,k}).}$$

Equivalently, for every positive divisor $r \mid k$ such that $k \geq mr^2 - r$, the pair

$$\left(mr, \frac{k}{r} + 1\right)$$

is a genuine globally Pareto-minimal point of $B_{m,k}$.

Proof. The canonical source recoding gives a recognizer using $m\lceil k/(g-1) \rceil = mr$ states and g stack letters. Here $C = mk + 1 = mr(g-1) + 1$. Writing $q = mr$, we have

$$q(g-1) = C - 1 < C, \qquad (q-1)g = qg - g < C$$

because $g \geq q$. Lemma 3.2 therefore proves global Pareto minimality. The divisor formula follows by putting $g = k/r + 1$ and rearranging $g \geq mr$. □

Corollary 6.3 (Arbitrarily many explicitly located points)**.** *For each fixed $m \geq 2$ and $t \geq 1$ there exist infinitely many k with at least t* explicitly located*, pairwise distinct elements of* $\mathrm{Pareto}(B_{m,k})$*.*

Proof. Choose k divisible by $\mathrm{lcm}(1, \ldots, t)$ and with $k \geq mt^2$. Every $r \in \{1, \ldots, t\}$ divides k and satisfies $k \geq mr^2 - r$, so Theorem 6.2 applies. As r increases, the state count mr increases and the stack-letter count $k/r + 1$ decreases. □

Remark 6.4 (Exact points versus exact thresholds)**.** *An equality $a_g = q$ does not by itself imply (q, g) is Pareto-minimal: there may be a plateau $a_{g-1} = a_g$. Theorem 6.2 proves both-coordinate minimality, while Theorem 6.1 asserts the exact threshold without claiming every interior point is on the frontier.*

7 Uniform order of Pareto-frontier width

Theorem 7.1 (Uniform width bound)**.** *For every $m \geq 2, k \geq 1$, set $L = mk + 1$, $D = mk + \min(m, 3) - 1$ and $T = \lfloor \sqrt{L} \rfloor$. Then*

$$\boxed{\left\lceil \frac{T}{2} \right\rceil \leq |\,\mathrm{Pareto}(B_{m,k})| \leq 2\left\lceil \sqrt{D} \right\rceil - 1.}$$

In particular $|\,\mathrm{Pareto}(B_{m,k})| = \Theta(\sqrt{mk})$ with absolute constants independent of m, k.

Proof. For the state axis, (1) shows that for $3 \le p \le \min(m, T)$,

$$b_{p-2} \ge \left\lceil \frac{L}{p-2} \right\rceil > 1 + \left\lceil \frac{D}{p} \right\rceil \ge b_p.$$

Indeed, $L \ge p^2$ and $\chi = \min(m, 3) \le 3$ imply $2L > (p-2)(2p+\chi-2)$ and thus $L/(p-2) > 2 + D/p$. The strict ceiling step uses $y > x+2 \Rightarrow \lceil y \rceil > 1 + \lceil x \rceil$; each disjoint two-position block must therefore contain at least one strict decrease and hence a distinct state-wing Pareto point. If $m \ge k$, then $T \le m$, so this gives at least $\lceil T/2 \rceil$ points including the $q = 1$ endpoint.
For the alphabet axis, let $4 \le g \le \min(k, T)$. From (2),

$$\frac{L}{g-2} - \frac{D}{g-1} = \frac{L - (\chi-2)(g-2)}{(g-2)(g-1)} > 1,$$

using $g^2 \le L$ and $\chi \le 3$. Hence $a_{g-2} > a_g$. Together with the attainable $g = 2$ endpoint and Lemma 3.3, this gives at least $\lfloor \min(k, T)/2 \rfloor$ distinct alphabet-wing points. When $k \ge m$, $T \le k$, and the $q = 1$ endpoint lies outside this wing because its alphabet budget exceeds k. Therefore $|\operatorname{Pareto}(B_{m,k})| \ge 1 + \lfloor T/2 \rfloor \ge \lceil T/2 \rceil$.
For the upper bound write $t = \lceil \sqrt{D} \rceil$. Every Pareto point belongs to the state wing $q \le m$ or the alphabet wing $2 \le g \le k$, since the canonical pair $(m, k+1)$ dominates any pair strictly outside these wings. If $t \le k-1$, count at most t possible strict alphabet thresholds at $g = 2, \ldots, t+1$. Every subsequent strict drop consumes at least one unit of a_g, and therefore there can be at most $a_{t+1} - a_k$ additional drops. The state wing has at most m points, while $a_k \ge m+1$ by the product bound. It follows that

$$|\operatorname{Pareto}(B_{m,k})| \le t + a_{t+1} - a_k + m \le t + \left\lceil \frac{D}{t} \right\rceil - 1 \le 2t - 1.$$

If instead $t \le m-1$, the symmetric state-wing argument gives at most $t + b_t - b_m$ state-wing points; using $b_t \le 1 + \lceil D/t \rceil$ and $b_m = k+1$, and allowing at most $k-1$ alphabet-wing points, we again obtain at most $2t-1$. Otherwise $t \ge m, k$, and the trivial $m + k - 1 \le 2t - 1$ bound suffices. The three possibilities are exhaustive: either $t \le k-1$, or $t \le m-1$, or both fail, in which case the integer t satisfies $t \ge k$ and $t \ge m$. This covers every parameter pair. □

8 Full state-coordinate support in a square-root wedge

Our uniform cardinality theorem does not say which state budgets support a minimal pair. The next theorem gives a complete state-*coordinate* support region, while deliberately making no claim to know the corresponding numeric stack minima b_p.

Theorem 8.1 (Every state coordinate is supported)**.** *Let $m \ge 4$ and put*

$$K_m = 2m + 2 - \left\lceil \sqrt{8(m+1)} \right\rceil.$$

If $k \ge K_m$, then

$$b_1(B_{m,k}) > b_2(B_{m,k}) > \cdots > b_m(B_{m,k}) = k+1.$$

Consequently the m pairs (p, b_p) for $p = 1, \ldots, m$ are all members of the full unrestricted Pareto frontier.

Proof. Write $L = mk + 1$, $N = mk + 2$, and $\chi = \min(m, 3) = 3$. The product lower bound and the one-state-source compiler provide

$$b_{p-1} \ge \left\lceil \frac{L}{p-1} \right\rceil, \qquad b_p \le 1 + \left\lceil \frac{N}{p} \right\rceil \quad (2 \le p < m),$$

while the canonical source realizes $b_m = k+1$. It is therefore sufficient to verify

$$\left\lceil \frac{L}{p-1} \right\rceil > 1 + \left\lceil \frac{N}{p} \right\rceil \quad (2 \le p < m), \qquad \left\lceil \frac{L}{m-1} \right\rceil > k+1. \tag{3}$$

If $k \ge 2m-4$, then for $2 \le p < m$,

$$L \ge m(2m-4)+1 \ge (p-1)(2p+1).$$

Thus $L/(p-1) - N/p = [L-(p-1)]/[p(p-1)] \ge 2$, and the first test follows. Moreover $L-(m-1)(k+1) = k-m+2 > 0$, so the last test follows.
It remains to consider $K_m \le k \le 2m-5$. Set

$$c = 2m-k, \quad d = \lceil \sqrt{8(m+1)} \rceil, \quad c_* = d-2.$$

Then $5 \le c \le c_*$ and $(c+1)^2 \le (d-1)^2 < 8(m+1)$. For $2 \le p < m$ set $a = m-p+1$ ($2 \le a \le m-1$). Direct division gives

$$\frac{L}{p-1} = 2m+2a-c+\frac{R(a)}{m-a}, \qquad R(a) = 2a^2 - ca + 1,$$
$$\frac{N}{p} = 2m+2a-c-2+\frac{U(a-1)}{m-a+1}, \qquad U(b) = 2b^2 - cb + 2.$$

Therefore the first test in (3) is equivalent to

$$\left\lceil \frac{R(a)}{m-a} \right\rceil \ge \left\lceil \frac{U(a-1)}{m-a+1} \right\rceil. \tag{4}$$

By completing the square,

$$R(a) + (m-a) = 2a^2 - (c+1)a + (m+1) \ge m+1 - \frac{(c+1)^2}{8} > 0,$$

so the left-hand ceiling in (4) is at least zero. If $U(a-1) \le 0$, the right-hand ceiling is at most zero. Otherwise put $b = a-1 \ge 1$. The inequality $U(b) > 0$ implies $b > (c-2)/2$, since otherwise $U(b) \le -2b+2 \le 0$. Hence $a > c/2$ and

$$R(a) - U(a-1) = 4a - c - 3 > c - 3 \ge 2.$$

The numerator $R(a)$ is strictly larger than the positive numerator $U(a-1)$, while its denominator is smaller, so (4) again follows. Finally $c \le c_* \le m+1$ gives $L-(m-1)(k+1) = m-c+2 > 0$. All comparisons hold, and each strict drop yields a Pareto point by coordinatewise minimality. □

The threshold above is not asserted to be the earliest k at which the *actual* DPDA state wing becomes fully supported. The following optimality statement concerns exactly the numerical sufficient comparisons used in the proof, a distinction important for avoiding unwarranted optimality claims.

Proposition 8.2 (Exact threshold for this arithmetic certificate)**.** *For each $m \ge 4$, the conjunction of all inequalities in* (3) *holds if and only if $k \ge K_m$.*

Proof. The forward direction for $k \ge K_m$ was proved above. Suppose $k < K_m$ and put $c = 2m-k$. If $k \le m-2$, then $L-(m-1)(k+1) = k-m+2 \le 0$, so the last test fails. We may therefore assume $m-1 \le k < K_m$, equivalently $c_*+1 \le c \le m+1$.
Let $n = 8(m+1)$ and $d = \lceil \sqrt{n} \rceil = c_*+2$. Choose a_0 to be a nearest integer to $d/4$; for $m \ge 4$ one has $2 \le a_0 \le m-1$. Put $e = 4a_0 - d$. The value $e^2 \in \{0,1,4\}$ is the least nonnegative integer congruent to d^2 modulo 8: taking a_0 nearest to $d/4$ makes $|4a_0 - d|$ minimal, and the square of an integer congruent to d modulo 4 has this least residue. Since $d^2 - n \ge 0$ and n is divisible by 8, we have $d^2 - n \ge e^2$, and consequently

$$8(2a_0^2 - da_0 + m + 1) = e^2 - (d^2 - n) \le 0.$$

Define $T_c(a) = 2a^2 - (c+1)a + m + 1$. As $c + 1 \geq d$, the preceding inequality gives $T_c(a_0) \leq 0$, whereas $T_c(1) = m + 2 - c \geq 1$. Choose the first integer $a \in \{2, \ldots, a_0\}$ with $T_c(a) \leq 0$. Then $T_c(a-1) > 0$, and

$$R(a) + (m - a) = T_c(a) \leq 0, \qquad U(a-1) + (m - a + 1) = T_c(a-1) + 1 > 0.$$

Thus the left ceiling in (4) is at most -1 and the right ceiling at least 0, so the corresponding test fails. This proves the equivalence. □

Corollary 8.3 (Sharper width on the supported-state region)**.** *For $m \geq 4$, $k \geq K_m$, and $T = \lfloor \sqrt{mk+1} \rfloor$,*

$$|\operatorname{Pareto}(B_{m,k})| \geq m + \left\lfloor \frac{\min(k, T)}{2} \right\rfloor.$$

Proof. Theorem 8.1 provides m distinct Pareto minima with at most m states and hence at least $k+1$ stack letters. The alphabet-wing two-step argument in the proof of Theorem 7.1, together with the unary-alphabet impossibility lemma, supplies $\lfloor \min(k, T)/2 \rfloor$ distinct minima with at most k stack letters. The two groups are disjoint. □

Remark 8.4 (Weaker and stronger numerical certificates)**.** *The sufficient certificate established here uses only $qg \geq mk + 1$. A stronger product bound, if separately established, may yield further numerical Pareto points; no such strengthening is used in this paper.*

Remark 8.5 (Model-dependent literature comparison)**.** *The classical recodings cited in the introduction do not directly state the exact corner theorem or the uniform frontier-width theorem for the concrete $B_{m,k}$ witness. The present results depend on the precise deterministic and simultaneous final-and-empty model. We do not claim novelty for generic state–stack alphabet recodings.*

9 Illustrative exact points and scope

For $B_{3,60}$, Theorem 6.2 certifies $(3, 61)$, $(6, 31)$, $(9, 21)$, and $(12, 16)$. For $B_{5,120}$ it certifies $(5, 121)$, $(10, 61)$, $(15, 41)$, $(20, 31)$, and $(25, 25)$. These are subsets of the full frontier, not complete coordinate lists. Theorem 8.1 also guarantees full state-coordinate support for $B_{12,15}$ and for $B_{100,173}$, implying respectively at least 18 and 165 Pareto points by Corollary 8.3; these are cardinality lower bounds, not complete coordinate sets.

The product inequality used here is $qg \geq mk + 1$. A potentially stronger product inequality would alter some small arithmetic corners, but is not required for the stated results and is not invoked.

10 Discussion and open problems

The exact all-parameter two-resource Pareto classification of $B_{m,k}$ remains open. The current results identify infinite exact subfamilies, prove the uniform cardinality order, certify every state coordinate in a large parameter region, and specify some exact thresholds. The conventional state-alphabet exchange mechanisms used constructively here are not asserted to be new. The 1982 and 1993 state/stack conversion papers and the 2020 complexity study are central comparators. The exact coordinates of every Pareto point for arbitrary (m, k) remain unknown. Related open questions include the complete characterization of plateaux in a_g and b_q, as well as extensions to restricted replacement lengths.

Statements and Declarations

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Code availability. The mathematical arguments establishing the stated general results are provided in the article. Auxiliary exact-integer comparisons and finite-word checks were used to cross-check selected examples and numerical inequalities; they are not needed as additional premises for those mathematical arguments. No verification software, executable certificate, dataset, or supplementary code file is included with this manuscript. The author-profile GitHub address does not identify a public repository for these checks.

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